

CloudPeak SaaS Solution Guide

CloudPeak SaaS Subscription & Churn Dashboard — Full Solution & Explanation Guide

Every step, every formula, broken down piece by piece — including WHY each decision was made

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Project Overview

Dataset: CloudPeak_SaaS_RAW.xlsx — 423 raw rows of subscription data, intentionally messy. **Final verified numbers:** - Total Customers: 400 - Active Customers: 289 - Churned Customers: 111 - Churn Rate: 28% - Total MRR: \$55,740 (approx.) - ARR (Annualized): \$668,840 (approx.) - ARPA: \$192.86

Purpose of This Project — What We Were Trying to Find

The scenario, in plain terms: CloudPeak is a made-up subscription company with customers on 3 different pricing plans across 4 regions. Leadership doesn't just want to know "how much money are we making" — they want to know where that money is coming from, which customers are most likely to cancel, and whether their best-paying customers are also their riskiest ones.

The simple questions we wanted answered: 1. How much recurring money is the company actually making right now? 2. What percent of customers cancel, and does that change by plan? 3. Which region brings in the most money? 4. Are the highest-paying customers all on one plan — and is that risky?

What we actually found: - The company makes about **\$55,740 a month** in recurring revenue, which works out to roughly **\$668,840 a year**. - **28% of customers cancel** — a pretty high number for this kind of business. - Cancellation isn't the same everywhere:

Enterprise customers cancel the most (33%), while Professional-plan customers cancel the least (24%). - Money comes in fairly evenly across all 4 regions, with North America only slightly ahead. - The biggest finding: **every single one of the top 10 highest-paying customers is on the Enterprise plan** — meaning the plan that makes the most money is also the plan most likely to walk away. That's a real risk worth flagging.

Why this project is worth showing off: it shows getting a tricky detail right — only counting CURRENTLY paying customers toward monthly revenue, not old customers who already left (a mistake beginners often make). It also shows connecting two separate charts into one bigger insight, instead of just reading off numbers one at a time.

PHASE 0: The Audit — What Was Looked For and Why

- Where the real table starts (title/merged rows push headers down)
 - Plan and Region — same real value spelled differently
 - MRR stored as text mixed with clean numbers (\$, /mo, USD)
 - Negative MRR values (impossible at face value — needed investigation)
 - Meaningful blanks (Seats, Support Tickets, NPS Score — not every account has full data)
 - Duplicate and blank rows
-

PHASE 2: Cleaning — Full List with Reasoning

Column	Fix Applied	Why
Header rows	Remove Top Rows (2), Use First Row as Headers	Title/spacer row pushed real headers down
Plan	Replace Values per spelling variant → 3 clean values	"basic"/"BASIC"/"Basic" would fragment into false-separate categories
Region	Replace Values (NA→North America, EU→Europe, etc.) → 4 clean values Change Type→Text	Acronym/full-name mismatch would break any region-based chart

MRR	(undo auto-error) → Replace Values (\$ /mo USD → nothing) → Change Type→Decimal Number	Classic “fix value before type” pattern
CSM Owner	Trim + Capitalize Each Word	Removes invisible whitespace/casing inconsistency
Signup Date / Churn Date	Cleaned to Date type; Churn Date blanks preserved	Blank Churn Date = still active customer — meaningful, not missing
Seats / Support Tickets / NPS Score	Left untouched, blanks preserved	Not every account has this data tracked or a survey response — filling with 0 would corrupt any AVERAGE()/SUM() logic later
Blank rows	Filtered via CustomerID → Remove Empty	Catches rows even after later custom columns are added
Duplicates	Select all columns → Remove Duplicates	Ensures only TRUE full- row duplicates removed
Negative MRR (judgment call)	Kept as-is; optional MRR Type flag column added	Represents real refunds/credits — correctly reduces total MRR when summed, rather than being fabricated away

Result: 400 clean rows, verified against DISTINCTCOUNT and filter dropdown checks.

PHASE 3: Data Modeling — Same Reasoning Pattern as HR, Different Entities

Tables built:

1. **Subscriptions** (fact table — one row per customer account)
2. **Plans** (dimension table)
3. **Calendar** (dimension table)

Why no separate “Customers” table this time

Unlike NorthStar’s Orders (where one customer could place MANY separate order rows, creating genuine repetition to extract), each row in Subscriptions already represents ONE customer’s ONE current subscription. There’s no multiple-rows-per-customer pattern here to de-duplicate — so, same reasoning as the HR project, Subscriptions itself serves as both fact and (only) customer-level dimension.

Why Plans still got its own table

Plan names (“Basic”, “Pro”, “Enterprise”) repeat across MANY customer rows — many customers share the same plan. This DOES fit the repetition pattern, so extracting it into a small dimension table was worthwhile, consistent with the general rule: **extract a dimension only when the same value repeats across multiple rows of your fact table.**

Quick Reference: Why These 2 Extra Tables (Plans + Calendar)

Table	Extracted because...	Row count check
Plans	The value repeats across MANY subscription rows (multiple customers share “Basic”, “Pro”, “Enterprise”) — repetition is exactly what makes a value worth pulling into its own dimension table	3 rows, verified against DISTINCTCOUNT('Subscriptions' [Plan])
Calendar	Not extracted from existing data at all — GENERATED fresh to guarantee a complete, unbroken date range, which SUMMARIZE on Signup Date alone could never provide (it would skip any day with zero signups)	Full Jan 2022-Dec 2024 range, one row per day

Same one-line rule as the HR project: extract a table when a value repeats across rows (Plans) OR when you need guaranteed date completeness for time-based analysis (Calendar). Subscriptions itself needed neither treatment — see below for why no separate Customers table was built.

Why Calendar connects to Signup Date, not Churn Date

Only one relationship between two tables can be ACTIVE at a time. This dashboard’s core story is about **acquisition and revenue growth over time** (MRR trend, signups by month) rather than primarily “when did people leave” — so Signup Date was chosen as the active connection, consistent with the dashboard’s main purpose.

PHASE 4: DAX Measures — Full Breakdowns

Why these 7 measures specifically

Each maps to a real SaaS business question:

Measure	Business Question
Total Customers	“How many accounts have we

	ever had?"
Active Customers	"How many are paying us right now?"
Churned Customers	"How many have we lost?"
Churn Rate	"What % of customers leave — the core SaaS health metric"
Total MRR	"How much recurring revenue do we have RIGHT NOW?"
ARR	"What's our revenue run-rate on an annual basis?"
ARPA	"How much does the average paying account contribute?"

Measure 1-3: Customer Counts

Total Customers = `DISTINCTCOUNT('Subscriptions'[CustomerID])`

Active Customers =
`CALCULATE(DISTINCTCOUNT('Subscriptions'[CustomerID]),`
`'Subscriptions'[Status] = "Active")`

Churned Customers =
`CALCULATE(DISTINCTCOUNT('Subscriptions'[CustomerID]),`
`'Subscriptions'[Status] = "Churned")`

Same CALCULATE + conditional filter pattern used in the HR project's Total Attritions measure — DAX has no native "COUNT IF," so CALCULATE with a condition is the standard substitute.

Measure 4: Churn Rate

Churn Rate = `DIVIDE([Churned Customers], [Total Customers], 0)`

Same DIVIDE-with-zero-fallback pattern as every rate measure so far — protects against a blank/error if a filtered view produces zero total customers.

Measure 5: Total MRR — the most important formula in this project

Total MRR =
`CALCULATE(SUM('Subscriptions'[MRR]), 'Subscriptions'[Status] = "Active")`

Piece by piece: - `SUM('Subscriptions'[MRR])` — would normally add up EVERY row's MRR value, including churned customers -
`CALCULATE(..., 'Subscriptions'[Status] = "Active")` — restricts that sum to ONLY rows where Status = "Active"

Why this restriction matters — a common real-world SaaS reporting mistake: A churned customer's row still physically EXISTS in your data with their last-known MRR value sitting in it. If you just ran `SUM('Subscriptions'[MRR])` with no filter, you would be counting revenue from customers who no longer pay you — silently overstating your actual current recurring revenue. This is a genuinely common mistake even among real analysts new to SaaS metrics, and this measure's CALCULATE filter is what prevents it.

Measure 6: ARR

ARR (Annualized) = `[Total MRR] * 12`

The simplest measure in the project — MRR is a MONTHLY figure; multiplying by 12 projects it forward as an annual figure. No CALCULATE needed since it's just referencing another measure and doing simple multiplication.

Measure 7: ARPA

ARPA = DIVIDE([Total MRR], [Active Customers], 0)

Average Revenue Per Account — how much the typical paying customer contributes. Notice this measure references TWO other measures ([Total MRR] and [Active Customers]) rather than recalculating from raw columns — this keeps the logic consistent and centralized: if the definition of “Active” ever changed, both this measure and Total MRR would automatically stay correct without needing separate updates.

Why We Picked These Exact Charts

- One important number (Total MRR, Churn Rate) → **Card**. A single fact, so just make it big and clear.
- Money coming in over time → **Line chart**. Best way to show something rising or falling.
- Cancel rate across different plans → **Bar chart, biggest to smallest**. Easiest way to compare categories.
- Money coming in across different regions → **Bar chart, biggest to smallest**. Same reason.
- Finding the specific highest-paying customers → **Table with a Top 10 filter**. This is a “who exactly” question, and a table with real names does that better than any chart.

PHASE 5: Report Build — What Was Built, What Went Wrong, and Why

Final report structure

- Title/Subtitle, 5 KPI Cards (Active Customers, ARR, Churn Rate, ARPA, Total MRR)
- Line chart: MRR by Signup Month
- Bar chart: Churn Rate by Plan (sorted descending)
- Bar chart: Total MRR by Region (sorted descending)
- Table: Top 10 customers by MRR
- Slicers: Plan, Region, Status
- Insight callout + credit line

Issue 1: Top N table filter — using the wrong filter card

What happened: While setting up the Top 10 Customers table, a filter dropdown labeled “Sum of MRR” with options like “is less than,” “is greater than” was opened first — this is a basic VALUE filter, not a Top N ranking filter, and using it would have required manually guessing a dollar threshold instead of cleanly asking for “the top 10.”

The correct control: In this version of Power BI, Top N filtering is done through a card labeled “**Data limit**” in the visual’s filter pane — entering a number there (10) and specifying the ranking field achieves the actual “top N by value” behavior, rather than a manual greater-than/less-than threshold.

The general lesson: Power BI's filter pane can contain multiple different filter TYPES for the same field (a basic value filter vs. a Top N/Data limit filter) — it's worth reading the filter card's label carefully rather than assuming the first one you see is the right tool, especially since their names have changed across Power BI versions.

Issue 2: The MRR trend line — signups vs true recurring revenue

What was noticed: The “MRR by Month” line chart rises through the first half of the year and declines toward December — this reflects SUM(MRR) grouped by SIGNUP month, meaning it's really showing **how much new MRR was added by companies who signed up in each month**, not a true month-over-month running total of recurring revenue (which would need to account for customers who signed up in an earlier month and are still paying in a later month).

Why this distinction matters: A true “MRR growth” chart in a real SaaS company would show a cumulative or point-in-time snapshot each month (accounting for ALL currently-active subscriptions at that point in time, regardless of signup date) — not just new signups grouped by month. Since building a true point-in-time MRR snapshot requires more advanced time-intelligence DAX than this project's core measure set, the chart was deliberately retitled to **“MRR by Signup Month”** instead of a generic “MRR Trend” — an honest label matching exactly what the chart actually calculates, rather than implying a more sophisticated metric than what was built.

The lesson: Always make sure a chart's title accurately describes what it's mathematically doing — a mismatch between title and actual calculation (even if unintentional) can misrepresent a finding to someone reading the dashboard.

Final verified numbers from the built report

Chart	Finding
Churn Rate by Plan	Enterprise highest (33%), Professional lowest (24%)
Total MRR by Region	North America leads, but spread across all 4 regions is relatively even
Top 10 Customers by MRR	Every single one of the top 10 accounts is on the Enterprise plan

Why the insight callout was worded the way it was

The chosen callout — *“All of CloudPeak's top 10 highest-value accounts are on the Enterprise plan — reinforcing Enterprise as the core revenue driver despite having the highest churn rate”* — deliberately surfaces a genuinely interesting tension in the data: Enterprise is simultaneously the MOST valuable segment AND the segment most at risk of churning. This is a more useful, nuanced insight than either fact stated alone, and demonstrates connecting two separate charts (the Top 10 table and the Churn Rate by Plan chart) into one combined finding — a stronger analytical move than simply restating one chart's headline number.

Key Takeaways From This Project

1. **The same “does this entity repeat across rows” test from the HR project applies here too** — Subscriptions didn't need a separate Customers table for the same underlying reason HR

Export didn't: each row already represents one unique account.

2. **Always filter recurring-revenue metrics to ACTIVE records only** — summing a status-agnostic column (like MRR) without restricting to currently-active rows is a common, serious SaaS-reporting mistake that silently overstates current revenue.
3. **A measure can safely reference other measures** — ARPA referencing [Total MRR] and [Active Customers] keeps your model's logic centralized and consistent, rather than recalculating raw definitions in multiple places.
4. **Power BI's filter pane can have multiple different filter types for the same field** — check the actual filter card's label (e.g., "Data limit" for Top N) rather than assuming the first filter option you see is the correct tool.
5. **A chart's title should accurately describe what it's mathematically calculating** — "MRR by Signup Month" is an honest label for a signup-grouped sum; a generic "MRR Trend" title would have implied a more sophisticated (and not-yet-built) point-in-time revenue snapshot.
6. **The strongest insights often come from connecting two separate charts**, not just restating one chart's top result — the Enterprise-is-valuable-AND-highest-churn finding required looking at the Top 10 table and the Churn Rate by Plan chart together.